

Information Security Management System for GreenGrid Energy

Master's in Computer Engineering - Cybersecurity and Systems
Administration
Security Management Systems (SIGSEG)

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Acronyms

APTs	Advanced Persistent Threats
AWS	Amazon Web Services
CEO	Chief Executive Officer
CFO	Chief Financial Officer
CIO	Chief Information Officer
CISO	Chief Information Security Officer
CSF	Cybersecurity Framework
ICS	Incident Command System
COO	Chief Operating Officer
CIS	Center for Internet Security
CTO	Chief Technology Officer
DER	Distributed Energy Resources
ERP	Enterprise Resource Planning
HR	Human Resources
ISMS	Information Security Management System
IT	Information Technology
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
IoT	Internet of Things
NIST	National Institute of Standards and Technology
NIS2	Network and Information Systems Directive II
OT	Operational Technology
PII	Personally Identifiable Information

RD	Research & Development
SCADA	Supervisory Control and Data Acquisition
CRM	Customer Relationship Management
LAN	Local Area Network
OT	Operational Technology
VPN	Virtual Private Network
AD	Active Directory
MFA	Multi-factor Authentication
RFPs	Requests For Proposals
RBAC	Role Based Access Control
RFQs	Requests For Quotes
SLA	Service-Level Agreement
MSAs	Master Service Agreements
SOWs	Statements Of Work
SOC	Security Operations Center
SIEM	Security Information and Event Management
EU	European Union
IRP	Incident Response Plan
SAP	Systems Applications and Products

Chapter 1

GreenGrid Energy: Organizational Overview

1.1 Introduction

GreenGrid Energy is a large-scale enterprise³ specialized in renewable energy, with particular expertise in smart grid technology. The company employs approximately 400 professionals and has achieved revenues of around 35 million euros in the latest fiscal year. Its business activities are structured around three core revenue streams:

1. Smart grid infrastructure deployment (45%),
2. Renewable energy supply contracts (40%),
3. Energy monitoring and analytics software (15%).

With the expanding adoption of renewable energy solutions and increasing integration of Internet of Things (IoT) devices, GreenGrid Energy operates at the intersection of Operational Technology (OT) and IT. This convergence necessitates a comprehensive cybersecurity approach, robust governance frameworks, and specialized departmental oversight.

To effectively respond to these challenges, GreenGrid Energy has assembled a dedicated project team to assess and strengthen information security practices within critical departments, while collaboratively addressing overall cybersecurity objectives.

1.1.1 Project Team Structure and Individual Responsibilities

The project team is composed of three specialists, each focusing on securing a critical area of operation:

Research & Development

Inês Ribeiro (1201346@isep.ipp.pt) assesses and enhances security practices within the Research & Development (RD) department, safeguarding proprietary technologies, algorithms, and intellectual property. She ensures secure collaborative environments for technology pilot programs, mitigating risks from industrial espionage and cyber threats.

Supply Chain Management

Manuela Leite (1200720@isep.ipp.pt) is responsible for analyzing procurement processes, supplier security standards, and component lifecycle management. Her primary focus includes establishing guidelines to mitigate risks associated with third-party vendors, particularly regarding hardware integrity and supply-chain vulnerabilities.

Engineering & Operations

Eduardo Sousa (1200920@isep.ipp.pt) oversees the security and resilience of critical grid infrastructure, ensuring Supervisory Control and Data Acquisition (SCADA) systems, OT networks, and renewable energy integrations are safeguarded against cyber threats. His role includes maintaining compliance with industry regulations, implementing advanced monitoring solutions, and coordinating Incident Response Plan (IRP)s to minimize operational disruptions.

1.1.2 Collective Responsibilities

Beyond their individual roles, the team collaborates on key areas that affect multiple departments, including:

- **Engineering & Grid Operations:** Ensuring secure integration of emerging technologies, particularly within OT and IT networks, to maintain operational integrity.
- **HR:** Implementing Role Based Access Control (RBAC) and promoting a security-aware culture through employee training and awareness programs.
- **Legal & Administration:** Aligning departmental activities with legal and regulatory requirements, managing compliance frameworks, and streamlining incident reporting mechanisms.
- **Cybersecurity Strategy Development:** Creating comprehensive security policies, IRPs, and risk mitigation frameworks applicable organization-wide.

1.1.3 Forward-Looking Objectives

The project team's strategic objectives include:

1. Conducting comprehensive risk assessments to identify critical assets, vulnerabilities, and potential impacts across departments.
2. Implementing an Information Security Management System (ISMS) aligned with recognized international frameworks (such as ISO 27001 or National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF)).
3. Ensuring secure convergence and seamless integration between OT and IT infrastructures.
4. Fostering a proactive security culture emphasizing continuous improvement and awareness throughout the organization.

1.2 Governance and Leadership

GreenGrid Energy follows a governance model that balances strategic oversight with efficient operations:

- **Board of Directors:** Define strategic vision, approves major investments, and ensures compliance with energy sector regulations.
- **Chief Executive Officer (CEO):** Oversees day-to-day operations and reports directly to the Board.
- **Executive Management Team:**
 - *Chief Financial Officer (CFO):* Handles budget planning, financial compliance, and resource allocation.
 - *Chief Technology Officer (CTO):* Determines technological direction, including research initiatives and product development.
 - *Chief Operating Officer (COO):* Manages core operations, covering engineering, supply chain, and service delivery.
 - *Chief Information Officer (CIO):* Aligns IT strategy with business objectives and oversees IT infrastructure.
 - *Chief Information Security Officer (CISO):* Develops cybersecurity strategies and enforces best practices across OT and IT domains.
- **Department Directors or Managers:** Lead their respective divisions and report to the relevant C-level executive.

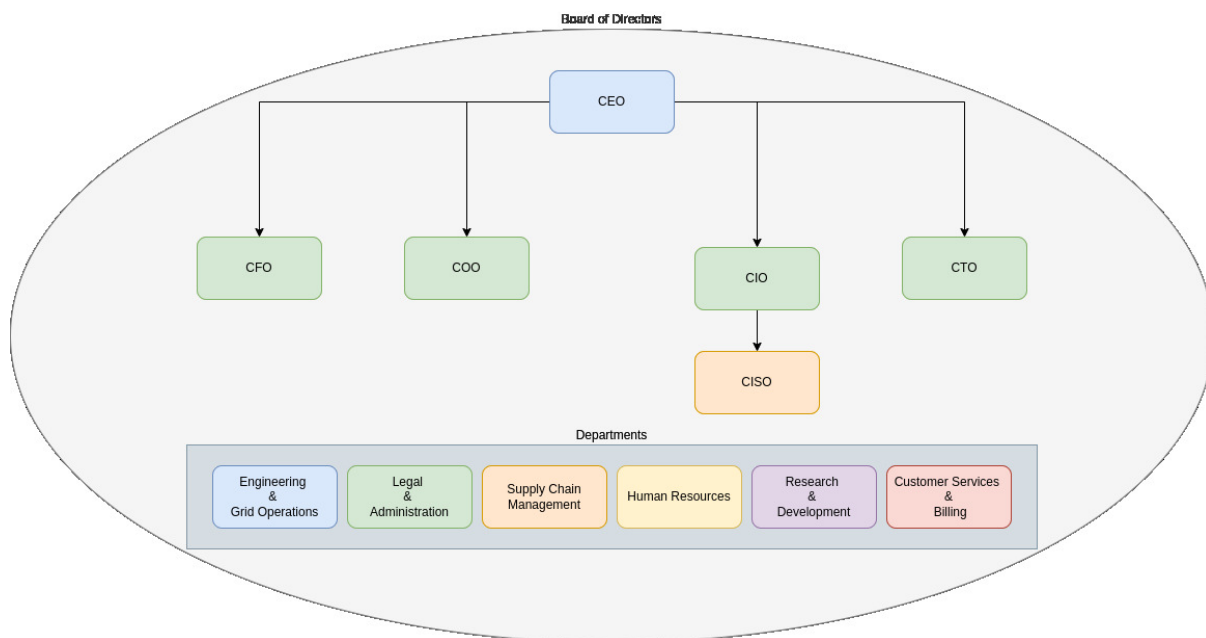


Figure 1.1: Organizational chart of GreenGrid Energy

Figure 1.1 provides an overview of the organization's structure. Each department has a dedicated director, while the executive team members focus on their respective areas of

expertise. Together with the CEO and department directors, these executives form the Board of Directors. All major decisions—including governance matters and the scope of new or revised policies—must be approved by this governing body before implementation throughout the company and its individual departments.

1.3 Key Departments

GreenGrid Energy employs a total of 400 staff members. Of these, 12 serve in leadership roles across the Board of Directors and C-level positions, leaving 388 employees distributed among the following six core departments. Each department is led by a Director or Manager, who reports to the relevant executive.

1.3.1 Engineering & Grid Operations

The Engineering & Grid Operations department at GreenGrid Energy is responsible for designing, maintaining, and optimizing the physical power distribution network. This includes real-time monitoring of energy flow, grid stability, and operational efficiency. The department plays a critical role in integrating renewable energy sources into the smart grid infrastructure while ensuring compliance with regulatory standards.

Core Responsibilities

The department's key responsibilities include:

- **Grid Infrastructure Management:** Overseeing the deployment, maintenance, and upgrades of smart grid components, including substations, transmission lines, and Distributed Energy Resources (DER).
- **SCADA System Operations:** Ensuring continuous monitoring and control of grid assets using SCADA systems to maintain grid stability and prevent outages.
- **Grid Security and Reliability:** Implementing resilience measures to protect against cybersecurity threats, natural disasters, and system failures.
- **OT Security:** Managing the convergence of IT and OT environments by deploying security controls for grid-connected devices and industrial control systems.
- **Incident Response and Disaster Recovery:** Developing contingency plans to mitigate risks associated with grid failures, cyberattacks, and supply chain disruptions.
- **Integration of Renewable Energy:** Managing the technical challenges of incorporating solar, wind, and other renewable energy sources into the smart grid.

Infrastructure and Technology Stack

The Engineering & Grid Operations department relies on the following core technologies:

- **SCADA Systems:** Used for real-time monitoring and control of critical grid components.

- **IoT Sensors and Smart Meters:** Providing detailed performance data from field assets to optimize energy distribution.
- **Edge Computing:** Enabling localized data processing for low-latency decision-making.
- **Hybrid Cloud Platforms (AWS/Azure):** Supporting predictive analytics, system automation, and secure data storage.
- **Firewall and Intrusion Detection System (IDS)/Intrusion Prevention System (IPS) Solutions:** Ensuring network security through robust threat detection mechanisms.
- **OT/IT Network Segmentation:** Implementing strict security controls to prevent unauthorized access between corporate IT and operational networks.

Departmental Management Structure

The Engineering & Grid Operations department is led by an **Engineering & Grid Operations Manager**, who reports directly to the COO. The internal structure is organized as follows:

- **Engineering & Grid Operations Manager:** Provides strategic oversight, ensures grid stability, and aligns department objectives with company goals.
- **Grid Infrastructure Lead:** Oversees grid modernization projects, substation maintenance, and infrastructure expansion efforts.
- **SCADA and Control Systems Specialist:** Manages SCADA operations, ensuring continuous monitoring, control, and security of grid systems.
- **OT Security Engineer:** Focuses on securing grid-connected assets and mitigating cyber threats targeting operational technology.
- **Renewable Energy Integration Specialist:** Ensures seamless incorporation of renewable energy sources, optimizing grid balancing and energy distribution.
- **Field Operations Supervisor:** Coordinates on-site maintenance teams, ensuring safety compliance and operational efficiency in power distribution.

1.3.2 Research & Development Department

Focused on innovation and next-generation solutions, the RD department is dedicated to advancing smart grid technology, predictive analytics, and renewable energy systems through proprietary innovations. It ensures secure environments for technology development and rigorously safeguards intellectual property against cybersecurity threats.

Organizational Structure

To address the responsibilities of the RD department, its management includes:

- **Director of RD:** Reports directly the developments to the Chief Technology Officer (CTO) and ensures that all research initiatives align with the overall strategic objectives.
- **Algorithm Development Specialists:** Design, develop, and optimize algorithms for smart grid optimization, predictive analytics, and renewable energy management.
- **Cybersecurity and Intellectual Property Management Specialists:** Safeguard proprietary research outputs, perform security assessments, and ensure proper protection of intellectual property.
- **Prototype Testing Specialists:** Conduct rigorous testing of prototypes in controlled environments to validate functionality, performance, and compliance with security standards.

Besides the organizational structure presented, there is a strong cross-department collaboration with Engineering & Grid Operations department, to facilitate the transition of validated technologies from the development stage to practical, operational environments.

1.3.3 Customer Services & Billing

All client-facing activities—such as account management, billing, and technical support—fall under this department’s purview, serving both residential and enterprise customers. Protecting personal and financial data is paramount, requiring rigorous compliance with data privacy regulations and prompt issue resolution to maintain customer satisfaction.

1.3.4 Supply Chain Management Department

The Supply Chain Management department at GreenGrid Energy plays a crucial role in supporting the company’s operational efficiency and strategic goals. Its primary responsibilities include procuring, assessing, and managing the acquisition of hardware, software, and external services necessary for the company’s operations. Given the company’s strong emphasis on cybersecurity and compliance, this department ensures that all acquired resources and services adhere to rigorous security standards and regulatory requirements.

To address the complexity of procurement and vendor management activities effectively, the Supply Chain Management department is organized with clearly defined roles under the supervision of a Supply Chain Manager, who directly reports to the COO. The internal structure of this department is as follows:

- **Supply Chain Manager:** Responsible for strategic oversight, departmental leadership, and alignment of procurement processes with organizational objectives. Coordinates overall procurement strategies and supervises team performance.
- **Hardware Procurement Specialist:** Oversees the procurement, logistics, inventory management, and verification of all physical hardware components, such as smart-grid devices, sensors, and power management equipment. Ensures hardware security integrity and vendor reliability.

- **Software Procurement Specialist:** Manages the procurement of software assets, licenses, analytics tools, and cybersecurity solutions. Collaborates closely with the CIO and CTO to confirm software solutions align with technology strategies and cybersecurity guidelines.
- **Service Procurement Specialist:** Focuses on acquiring external services critical to operations, such as consulting engagements, technical training programs, support contracts, and maintenance services. Responsible for monitoring the quality and compliance of service providers.
- **Vendor Relationship Specialist:** Dedicated to managing strategic vendor relationships across hardware, software, and services. Conducts vendor risk assessments, security audits, and compliance verifications to mitigate supply chain-related risks.

This structure ensures clear accountability, specialization of roles, and scalability to meet future operational demands. It also aligns closely with GreenGrid Energy’s cybersecurity framework by ensuring thorough vetting and continuous monitoring of all vendors and acquired resources.

The organizational hierarchy within the Supply Chain Management department is depicted below:

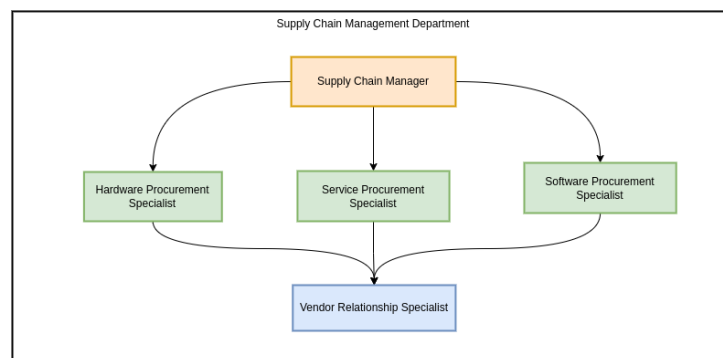


Figure 1.2: Organizational chart of Supply Chain Management Department

Through this structured organization, the Supply Chain Management department ensures efficient operations, robust security controls, and strategic alignment with GreenGrid Energy’s business objectives.

1.3.5 HR

HR oversees recruitment, training, performance evaluation, and overall employee welfare. As part of the company’s broader security posture, HR also manages background checks, role-based access control, and ongoing security awareness initiatives to ensure that all staff understand and adhere to established security protocols.

1.3.6 Legal & Administration

This department ensures that GreenGrid Energy remains compliant with evolving industry regulations and standards. Core tasks include contract drafting and review, regulatory monitoring, and oversight of administrative budgeting. By maintaining legal and procedural integrity, the Legal & Administration team supports the organization’s ability to operate smoothly and avoid compliance risks.

1.4 Infrastructure and Security Context

1.4.1 Core Technologies and Systems

GreenGrid Energy's infrastructure comprises several critical technologies supporting operational excellence, security, and departmental integration.

SCADA Systems These systems allow real-time control and monitoring of grid equipment, essential for the reliability and stability of energy distribution operations.

Hybrid Cloud Platforms (AWS/Azure) Offer scalable computing, secure data storage, advanced analytics, and support for IoT device management, facilitating innovation and departmental collaboration.

Edge Computing Provides local data processing capabilities near physical assets, ensuring low latency and rapid response for critical grid management tasks.

Microsoft 365 The primary platform for internal communication, collaboration, and productivity across all company departments.

Systems Applications and Products (SAP) Enterprise Resource Planning (ERP) Integrates business-critical functions such as procurement, inventory management, customer services, financial oversight, and regulatory compliance.

1.4.2 Network Topology

GreenGrid Energy maintains two distinct networks to segregate operational and corporate functions securely:

- **OT Network:** Hosts grid-specific operations, including SCADA systems and field-deployed IoT devices.
- **IT Network:** Manages corporate services such as ERP systems, collaboration tools, and HR applications.

Secure segmentation between the OT and IT networks is essential to safeguard critical infrastructure operations.

1.4.3 Threat Landscape

The primary cybersecurity threats facing GreenGrid Energy include:

- **Nation-State Actors:** Target critical infrastructure for strategic objectives and disruption.
- **Advanced Persistent Threats (APTs):** Sophisticated attackers leveraging vulnerabilities in IoT and grid-related technologies.
- **Supply Chain Risks:** Compromise of hardware or software components through third-party suppliers.

1.4.4 Technology and Department Association

The following table associates each core technology with its primary departmental users, highlighting responsibility and ensuring clear governance:

Table 1.1: Core Technologies and Associated Departments

Technology/System	Primary Department(s)
SCADA Systems	Engineering & Grid Operations
Hybrid Cloud Platforms (AWS/Azure)	IT, RD
Edge Computing	Engineering & Grid Operations, RD
Microsoft 365	All Departments
SAP ERP	Supply Chain, Customer Services, Legal & Administration, HR
OT Network	Engineering & Grid Operations
IT Network	IT, Legal & Administration, HR

This clear and organized mapping ensures effective oversight, accountability, and management of cybersecurity risks across the company's operational infrastructure.

Chapter 2

ISMS Scope

Aligned with ISO/IEC 27001 Clause 4.3 guidelines,² this chapter details GreenGrid Energy’s *ISMS* scope by identifying the key stakeholders, describing shared IT infrastructure, highlighting core business processes, and specifying the boundaries within which the ISMS will operate.

2.1 Organizational Stakeholders

GreenGrid Energy’s stakeholders, as defined by ISO/IEC 27001, include both internal and external entities that can influence—or be influenced by—the organization’s security practices.⁴ Table 2.1 summarizes the principal stakeholder groups and their roles in relation to the ISMS.

Table 2.1: Stakeholders and ISMS Interaction

Stakeholder Group	Interaction with the ISMS
Executive Management and Board	Provide strategic oversight, allocate resources, approve ISMS scope, and review security performance.
IT Department	Implement and maintain technical security controls; daily management of the ISMS infrastructure.
Department Heads and Employees	Comply with ISMS policies, handle information securely, report incidents, and participate in risk assessments.
Customers and Clients	Expect protection of data confidentiality, integrity, and availability as part of service agreements.
Suppliers and Third-Party Providers	Comply with contractual security requirements to ensure the security of GreenGrid data processed externally.
Regulators and Authorities	Expect adherence to industry regulations and may audit ISMS practices and compliance.
Certifying Bodies/Auditors	Evaluate and validate compliance with ISO/IEC 27001 through periodic certification audits.

2.2 Shared IT Infrastructure and Systems

GreenGrid Energy employs a centrally managed IT infrastructure used across multiple departments. Key elements of this shared environment include:

- **Network Infrastructure:** Company-wide network including Local Area Network (LAN), secure Wi-Fi, and Virtual Private Network (VPN) for remote employees.
- **Server and Cloud Systems:** Hosting internal applications, databases, and departmental services.
- **Enterprise Applications:** Microsoft 365, SAP ERP, and Customer Relationship Management (CRM) software supporting business operations and collaboration.
- **User Devices:** Managed desktop computers, laptops, and mobile devices.
- **Data Storage and Backup Systems:** Centralized data repositories and backup solutions.
- **Communication and Collaboration Tools:** Email systems, instant messaging platforms, video conferencing, and intranet portals.

Each department relies on these resources for day-to-day operations. Chapter 3 will focus on specific requirements, objectives, and policies around these systems, while this chapter captures their basic description for scope purposes.

2.3 Business Processes, Data, and Security Risks by Department

2.3.1 Engineering & Operations

The Engineering & Operations Department serves as the backbone of GreenGrid Energy's infrastructure, ensuring the seamless integration and reliable performance of smart-grid technologies and renewable energy systems. This department is responsible for designing, deploying, and maintaining critical operational assets, balancing efficiency, scalability, and cybersecurity. The ISMS scope encompasses key operational processes, addressing risks associated with grid stability, system automation, and industrial control systems while aligning with the organization's broader security objectives.

Grid Infrastructure Management:

- **Activities:** Overseeing deployment, maintenance, and modernization of grid components.
- **Data Processed:** Grid topology data, maintenance logs, infrastructure monitoring reports.
- **Cybersecurity Risks:** Unauthorized system modifications, physical and cyber tampering, outdated firmware vulnerabilities.
- **Technologies:** SCADA systems, asset management platforms, industrial-grade firewalls.

SCADA System Operations:

- **Activities:** Monitoring and controlling grid assets for stability and reliability.
- **Data Processed:** Real-time grid performance data, event logs, control commands.
- **Cybersecurity Risks:** Remote access exploitation, SCADA malware, data interception.
- **Technologies:** Secure remote access solutions, IDS, encrypted communication channels.

Incident Response and Disaster Recovery:

- **Activities:** Developing and executing contingency plans for cyber and operational incidents.
- **Data Processed:** Incident reports, forensic data, response logs.
- **Cybersecurity Risks:** Prolonged downtime, inadequate threat response, lack of forensic integrity.
- **Technologies:** Automated threat detection, backup and recovery solutions, Security Information and Event Management (SIEM) platforms.

Renewable Energy Integration:

- **Activities:** Managing the integration of solar, wind, and other renewable energy sources.
- **Data Processed:** Renewable generation data, demand-response metrics, battery storage status.
- **Cybersecurity Risks:** Grid imbalances, data manipulation, unauthorized control access.
- **Technologies:** Smart inverters, edge computing for energy optimization, blockchain for energy transactions.

2.3.2 Research & Development (R&D)

The RD Department plays a strategic role at GreenGrid Energy, providing innovative solutions essential for enhancing smart-grid operations and renewable energy technologies. The ISMS scope outlines the department's core processes, explicitly highlighting the data handled, relevant cybersecurity risks, and utilized technologies, aligning with the ISMS objectives.

Proprietary Research and Innovation Process

This process involves identifying, researching, and evaluating emerging technologies and concepts that can potentially enhance grid efficiency and reliability.

Activities:

- Market and technology research

- Feasibility analysis and experimentation
- Preliminary risk assessment of potential innovations

Data Processed:

- Market intelligence data
- Experimental research results

Cybersecurity Risks:

- Data leakage of research findings
- Unauthorized access to research environments

Technologies:

- Cloud platforms
- Research databases and analytics tools

Algorithm and Software Development

This process is dedicated to the systematic creation and validation of predictive analytics algorithms and software for operational use in grid optimization and energy distribution forecasting.

Activities:

- Specification of requirements
- Software coding and iterative development cycles
- Simulation-based testing and validation
- Continuous security assessments during development

Data Handled:

- Proprietary algorithmic code
- Simulated and historical grid data

Security Risks:

- Intellectual property theft
- Vulnerabilities in software code

Technologies:

- Advanced simulation tools
- Code repositories and CI/CD pipelines

Prototype Development and Testing

Prototypes of innovative solutions are built and securely tested to verify their effectiveness, reliability, and resilience to cyber threats.

Activities:

- Development of technology prototypes
- Secure environment configuration for testing
- Conducting controlled cybersecurity assessments
- Documentation and analysis of testing outcomes

Data Handled:

- Prototype technical specifications
- Test results and security assessments

Security Risks:

- Leakage of sensitive design specifications
- Unauthorized access during testing phases

Technologies Used:

- Testing and simulation environments
- Computing infrastructures

Intellectual Property Protection

Ensuring strong protection of intellectual property is essential for maintaining the competitive edge and minimizing risks of industrial espionage and data breaches.

Activities:

- Documentation and protection of R&D outputs
- Management of patent filing processes
- Regular security audits for IP management

Data Handled:

- Intellectual property records and documentation
- Sensitive research outcomes

Security Risks:

- Industrial espionage attempts
- Intellectual property data breaches

Technologies Used:

- Secure data storage solutions
- Encryption and secure communication tools

Technology Integration and Collaboration

This process focuses on transitioning validated technologies from the R&D department into operational use, working closely with Engineering & Grid Operations.

Activities:

- Collaborative planning and joint validation sessions
- Cross-departmental training and knowledge transfer
- Monitoring and support during initial deployment phases

Data Handled:

- Integration plans and documentation
- Performance and reliability data from integration tests

Security Risks:

- Integration-related vulnerabilities
- Compromise during the operational handover process

Technologies Used:

- Collaboration platforms (Microsoft 365, Jira)
- Secure configuration management systems

2.3.3 Customer Services

Key Processes and Data: Client contracts, billing data, CRM, customer Personally Identifiable Information (PII)

Security Risks: Privacy breaches, financial fraud, unauthorized data disclosure

2.3.4 Finance

Key Processes and Data: Financial management (budgeting, transactions, payroll, ERP financial module)

Security Risks: Financial fraud, phishing attacks, unauthorized access, data tampering

2.3.5 Supply Chain Management

Procurement and Sourcing

Oversees the end-to-end procurement lifecycle for hardware, software, and services by:

- Drafting Requests For Proposals (RFPs) or Requests For Quotes (RFQs).
- Conducting vendor evaluations and security due diligence.
- Negotiating contracts and maintaining supplier agreements.

Associated data supporting this process:

- Vendor profiles and security certifications.
- Product specifications (e.g., smart grid components, software licenses).
- Pricing details, purchase orders, and invoices.

Vendor Relationship Management

Manages ongoing relationships with approved suppliers to ensure adherence to GreenGrid Energy's security and performance standards by:

- Monitoring Service-Level Agreement (SLA) requirements.
- Tracking vendor performance and compliance metrics.
- Conducting periodic vendor risk assessments, especially for critical components.

Relevant data for this process:

- Contract terms and SLA metrics.
- Vendor incident reports or audit findings.
- Communication records (emails, project updates, meeting minutes).

Inventory & Logistics Coordination

Coordinates warehousing, transportation logistics, and inventory control once goods or services are procured by:

- Tracking shipment and delivery status of hardware components.
- Verifying integrity and authenticity of infrastructure parts.
- Overseeing storage, handling, and distribution to operational sites.

Data types maintained in this process:

- Shipping documentation (e.g., bills of lading, customs paperwork).
- Inventory levels and restocking details.
- Asset tracking logs (equipment serial numbers, batch numbers).

Compliance and Risk Management

Ensures that supplier relationships and procurement activities meet security policies and relevant regulations by:

- Reviewing supplier security audits and certifications.
- Ensuring compliance with data protection and trade regulations.
- Coordinating corrective actions for supplier or logistics risks.

Data associated with this process:

- Audit reports, compliance checklists, and regulatory filings.
- Documentation of risk treatment actions (vendor replacements, contract updates).
- Internal approval records and compliance sign-offs.

Contract and SLA Oversight

Manages contractual obligations from initial sign-off to renewal or termination by:

- Monitoring contract milestones (renewal dates, penalty clauses).
- Ensuring suppliers meet service quality and security requirements.
- Handling contract amendments for security or operational changes.

Key data involved:

- Master Service Agreements (MSAs), Statements Of Work (SOWs).
- Change requests and contract amendment records.
- Legal correspondence regarding contract disputes or terminations.

2.3.6 Human Resources

Key Processes and Data: Employee management (personal records, payroll data, performance data)

Security Risks: Unauthorized disclosure, insider threats, privacy breaches

2.3.7 IT Department

Key Processes and Data: Infrastructure management (network devices, servers, endpoint devices, IAM systems, incident response data)

Security Risks: System compromise, configuration errors, misuse of admin privileges

2.3.8 Sales and Marketing

Key Processes and Data: Customer relationship management (CRM), market data, public-facing web resources

Security Risks: Unauthorized access, reputation damage, phishing/malware attacks

2.4 ISMS Boundaries and Exclusions

The ISMS boundaries include:

- Internal IT and OT networks, including SCADA and enterprise systems used by the Supply Chain Management, Customer Services & Billing, and Research & Development departments.
- Data processing within cloud environments generated or consumed by Supply Chain Management, Customer Services & Billing, and Research & Development.
- Procurement processes and vendor management.
- Employee data, access management, and training processes.

Explicit exclusions from the ISMS scope include:

- Third-party systems fully managed externally, where GreenGrid has no direct control.
- Personal employee devices not centrally managed by the organization.
- Physical security at externally managed premises (e.g., leased data centers).
- Separate subsidiaries or partner organizations outside direct oversight.

These exclusions are justified by the lack of direct administrative oversight or practical manageability; any associated risks are mitigated through contractual obligations and vendor assessments.¹

Chapter 3

Information Security Policy and Objectives

This chapter addresses the next phase of establishing GreenGrid Energy's ISMS by incorporating stakeholder requirements, business goals, security objectives, leadership commitment, and overall compliance needs.

3.1 Stakeholder Requirements and Business Goals

As established in Chapter 2, GreenGrid Energy's stakeholders include Executive Management, the IT Department, Department Heads & Employees, Suppliers, Clients, Regulators, and Certifying Bodies. Because the ISMS scope specifically includes:

- Internal IT and OT networks, including SCADA and enterprise systems used by **Supply Chain Management, Customer Services & Billing**, and **RD** departments.
- Data processing within cloud environments generated or consumed by **Supply Chain Management, Customer Services & Billing**, and **RD**.
- Procurement processes and vendor management.
- Employee data, access management, and training processes.

Most stakeholder requirements focus on ensuring robust controls to protect:

- Critical infrastructure.
- Sensitive customer and financial data.
- Intellectual property from RD activities.
- Secure procurement of hardware/software through the Supply Chain Management department.
- Proper handling of employee data and permissions.

In contrast, stakeholders agree that **third-party systems fully managed externally, personal employee devices, physical security at externally managed premises, and separate subsidiaries or partner organizations outside direct oversight** are explicitly out of scope, thus reducing direct stakeholder interest in those areas.

3.2 Leadership Commitment

GreenGrid Energy's board of directors reinforce the ISMS by:

- **Allocating Sufficient Budgets:** Funding security tools and training for departments within the ISMS boundaries.
- **Championing Policy Adoption:** Ensuring all employees and department heads integrate the new Information Security Policy into their daily operations.
- **Encouraging a Security Culture:** Mandating ongoing training on secure processes, especially for staff involved in vendor management and grid operations.
- **Reviewing ISMS Performance:** Conducting periodic management reviews to track key risk metrics, incident reports, and audit findings.

3.3 Monitoring and Review of ISMS Performance

To ensure continuous improvement, GreenGrid Energy will measure and review ISMS effectiveness through:

- **Annual ISMS Audit:** Conduct full audits and ensure corrective actions for non-conformities.
- **Key Risk Indicators (KRIs):** Monitor unauthorized access attempts, security incidents, and policy violations.
- **Management Review Meetings:** Perform biannual ISMS performance reviews with Executive Management.

3.4 Risk Management Goals and Compliance Requirements

Given the potential threats to the included systems and data, the organization sets forth the following overarching risk management goals:

- **Protect OT and IT Convergence:** Keep SCADA, IoT, and enterprise networks resilient against cyber threats.
- **Safeguard Data Privacy and Integrity:** Ensure confidentiality of customer billing data, vendor records, and intellectual property from RD.
- **Maintain Regulatory Compliance:** Satisfy relevant European Union (EU) directives such as NIS 2, and meet ISO/IEC 27001 standards to secure continuous certification and avoid penalties.
- **Mitigate Vendor and Procurement Risks:** Enforce security checks during procurement of critical hardware/software, and conduct regular vendor risk assessments.
- **Enhance Incident Response Preparedness:** Develop rapid containment strategies for security breaches or supply chain disruptions, minimizing downtime of grid operations.

3.5 Information Security Policy (Clause 4.2)

Policy Statement

GreenGrid Energy is committed to safeguarding the confidentiality, integrity, and availability of information assets within the boundaries of its ISMS, specifically covering:

- Internal IT and OT networks that power core operations.
- Data processed through corporate cloud environments.
- Procurement and vendor relationships supporting operational continuity.
- Employee access management, training, and HR-related records.

Key Principles

- **Risk-Based Controls:** Security measures must be tailored to identified threats, focusing on high-risk areas such as SCADA, third-party supplier links, and RD data.
- **Legal and Regulatory Compliance:** Adhere to ISO/IEC 27001, Network and Information Systems Directive II (NIS2), and national data protection laws relevant to energy operations.
- **Continuous Improvement:** Policies, controls, and training programs will be reviewed at least annually or following major incidents.

Responsibilities

- **CISO:** Oversees policy implementation and audits across IT, OT, and supply chain operations.
- **Department Directors:** Ensure departmental processes adhere to this policy; sponsor secure practices among staff.
- **All Employees:** Follow approved procedures, maintain vigilance for potential threats, and report incidents promptly.

3.6 Department-Specific Security Objectives (Clause 6.2)

Below are the measurable information security objectives for each major department within the ISMS scope. Each subsection also highlights the primary stakeholders driving or benefiting from these goals.

3.6.1 Engineering & Operations

Stakeholders:

- **Grid Operations Teams:** Manage real-time grid stability and control systems.

- **SCADA and Industrial Control Engineers:** Maintain secure and reliable SCADA environments.
- **Regulatory Authorities:** Oversee compliance with national and EU grid security regulations.
- **IT Security and Network Teams:** Secure OT/IT convergence to protect grid operations.

Objectives:

- **Objective EO-1: Strengthen SCADA System Security**
 - **Target:** Ensure 100% of SCADA systems have up-to-date security patches and network segmentation controls.
 - **Measure:** Compliance rate of SCADA system updates and segmentation audits.
 - **Owner:** SCADA Security Team, CISO.
- **Objective EO-2: Improve OT Network Visibility and Anomaly Detection**
 - **Target:** Deploy real-time monitoring tools to achieve a 90% anomaly detection rate in OT networks.
 - **Measure:** Number of security alerts detected vs. actual threats mitigated.
 - **Owner:** Engineering Security Lead, IT Security Team.
- **Objective EO-3: Enhance Physical and Cyber Resilience of Grid Infrastructure**
 - **Target:** Perform biannual security assessments for grid substations and power distribution control centers.
 - **Measure:** Number of identified vulnerabilities mitigated before next assessment cycle.
 - **Owner:** Grid Operations Manager, CISO.

3.6.2 Supply Chain Management

Stakeholders:

- **Supply Chain Staff:** Procurement Specialists and Vendor Relationship Managers who oversee supplier performance and risk.
- **Vendors and Third Parties:** Partners required to comply with our stringent security checks, audits, and contractual requirements.
- **Executive Management:** Needs assurance of an uninterrupted, secure supply of components to support operational continuity and protect the company's reputation.

Objectives:

- **Objective SC-1: Reduce High-Risk Vendor Gaps**

To lower the risk of compromised components or software infiltrating GreenGrid Energy's infrastructure, it is essential that our critical suppliers adhere to rigorous security standards. This proactive auditing approach helps prevent operational disruptions, regulatory non-compliance, and reputational damage.

- **Target:** Audit at least 100% of all critical hardware/software suppliers annually.
- **Measurement:** Track the percentage of audited suppliers, monitor the number and closure time of security findings, and evaluate any security incidents attributable to supplier issues.
- **Owner:** Supply Chain Manager and CISO.

- **Objective SC-2: Maintain Rapid Supplier Incident Response**

Minimizing supply chain downtime is vital to avoiding customer dissatisfaction and churn. Quick resolution of disruptions ensures that delays do not negatively impact our operations or customer trust.

- **Target:** Ensure that supply chain incidents are resolved within a maximum of 45 minutes, except in cases of natural disasters or other uncontrollable events.
- **Measurement:** Monitor the average resolution time, count incidents exceeding the 45-minute threshold, and track customer feedback or churn linked to supply chain delays.
- **Owner:** Supply Chain Manager and CISO.

3.6.3 Research & Development (RD)

Stakeholders:

- **RD Teams:** Create proprietary analytics algorithms and software for grid optimization.
- **Executive Management:** Relies on RD innovation for market competitiveness.
- **Intellectual Property Owners:** Concerned about protecting patents, code repositories, trade secrets.

Objectives:

- **Objective RD-1: Prevent Unauthorized Access to RD Data** It is vital that the innovation data on GreenGrid Energy's Infrastructure remains unaccessed in a non authorized way, as such data includes confidential information that could cause reputational damage and eventual software damage on the infrastructure.

- **Target:** Achieve a 50% reduction in unauthorized access attempts and security alerts within the RD environment.
- **Measure:** Number of RD-specific security alerts against the prior reporting period.
- **Owner:** RD Manager, IT Department.

- **Objective RD-2: Strengthen Security of Environments** Since the environments on the RD Department are critical, attacks on its infrastructure are to be avoided. Strengthening security on the environments ensures the attacks and vulnerabilities don't affect the service provided.
 - **Target:** Perform quarterly audits of RD environments to identify and mitigate vulnerabilities.
 - **Measure:** Audit completion rate.
 - **Owner:** Director of RD IT Security Team.

Chapter 4

Other Security Policies and Procedures

In addition to our main Information Security Policy, GreenGrid Energy implements several additional organization-wide security policies and corresponding procedures. These measures support our compliance with ISO/IEC 27001 Annex A and ensure that all aspects of our security posture are addressed consistently across the organization.

4.1 Other Security Policies

4.1.1 Access Control Policy

Purpose: To ensure that access to information assets is granted only to authorized individuals in accordance with the principle of least privilege.

Scope: Applies to all employees, contractors, and third parties who access GreenGrid Energy's information systems.

Policy:

- Access rights are assigned based on roles and responsibilities.
- All access requests must be formally documented, approved, and periodically reviewed.
- Multi-factor Authentication (MFA) is required for accessing sensitive systems.

Enforcement: Failure to comply with access control procedures may result in restricted access, mandatory retraining, or disciplinary action, including termination if misuse is severe.

Responsibilities: The IT Department, in collaboration with HR and departmental managers, is responsible for the implementation and monitoring of access controls.

4.1.2 Data Classification Policy

Purpose: To classify information according to its sensitivity and ensure that it is handled appropriately to prevent unauthorized disclosure, modification, or destruction.

Scope: Applies to all data owned or processed by GreenGrid Energy.

Policy:

- Information shall be classified as Public, Internal, Confidential, or Restricted.

- Handling, storage, and transmission procedures will be defined based on the data classification.
- Regular reviews will ensure that classification levels remain accurate and effective.

Enforcement: Any violation of the data classification guidelines may lead to corrective measures such as access restrictions, formal warnings, and potential legal action in case of data breaches.

Responsibilities: Department heads are responsible for classifying information within their areas, while the IT Department ensures secure storage and transmission.

4.1.3 Incident Response Policy

Purpose: To establish a clear process for reporting, managing, and resolving security incidents efficiently.

Scope: All employees and third-party providers who use or manage GreenGrid Energy's information systems.

Policy:

- All security incidents must be reported immediately to the IT Security Team.
- An IRP is in place to contain, eradicate, and recover from incidents.
- A post-incident review is conducted to capture lessons learned and implement corrective measures.

Enforcement: Non-compliance with incident reporting procedures or failure to follow the response plan may result in disciplinary action, including termination, and can lead to further legal consequences if negligence results in significant harm.

Responsibilities: The IT Security Team, led by the CISO, is responsible for managing incident response and ensuring that all incidents are documented and reviewed.

4.1.4 Responsible IT Resource Usage Policy

Purpose: To promote the secure and efficient use of GreenGrid Energy's IT resources in support of business operations while maintaining data integrity.

Scope: This policy applies to all employees, contractors, and temporary staff who access or use GreenGrid Energy's IT systems and devices.

Policy:

- IT resources should primarily support work-related activities and organizational goals.
- Users must protect their access credentials and handle sensitive information responsibly.
- Personal use of IT resources is allowed only if it does not interfere with work duties or compromise system security.
- All IT resource usage must comply with applicable laws, regulations, and company policies.

Enforcement: Violations of this policy may lead to corrective measures such as access suspension, formal warnings, mandatory retraining, or disciplinary actions up to and including termination.

Responsibilities: Department Heads are responsible for ensuring that their teams understand and adhere to this policy, while the IT Department monitors usage and implements necessary safeguards.

4.2 Other Security Procedures

For each of the above policies, the following procedures provide detailed, step-by-step instructions for implementation and operational management. Beyond simply stating how tasks are carried out, these procedures also outline how data produced by or for the ISMS—such as audit logs, classification labels, or incident records—are securely stored, accessed, and disposed of.

4.2.1 Access Control Procedure

- **Document and Approve Access Requests:** All new access requests must be submitted via the official request form and approved by the department manager and IT Security.
- **Periodic Access Reviews:** Perform quarterly audits to ensure that user privileges align with current roles and responsibilities; remove or adjust access as needed.
- **MFA:** Enable and maintain MFA for sensitive systems, ensuring credentials are stored securely and never shared.
- **Protecting ISMS Data:** Access logs, approval workflows, and any reports on privileges are labeled *Restricted*, stored in an encrypted repository, and shared only with authorized personnel.

4.2.2 Data Classification Procedure

- **Identify and Label Assets:** Classify data as Public, Internal, Confidential, or Restricted according to the Data Classification Policy.
- **Storage and Handling Guidelines:** For each classification level, follow approved controls (e.g., encryption for Restricted data, restricted folder permissions for Confidential).
- **Periodic Review and Update:** Conduct semiannual checks to validate that classification labels remain accurate as business processes evolve.
- **Protecting ISMS Data:** All classification records and any logs describing data reclassification are stored on a secure server with role-based access. Any reclassification decisions are documented and version-controlled.

4.2.3 Incident Response Procedure

- **Immediate Reporting:** All security incidents must be reported to the IT Security Team or the Incident Response Coordinator within 15 minutes of discovery.
- **Containment, Eradication, and Recovery:** Follow the IRP to isolate affected systems, remove malicious artifacts, and restore operations with minimal downtime.
- **Post-Incident Review:** Conduct a root cause analysis to identify vulnerabilities or process gaps and document lessons learned for continuous improvement.
- **Protecting ISMS Data:** All incident reports, investigation notes, and evidence are tagged *Confidential*, stored in a secure incident database accessible only to the Incident Response Team, and retained for the legally required duration.

4.2.4 Responsible IT Resource Usage Procedure

- **Policy Distribution and Acknowledgment:** The Responsible IT Resource Usage Policy is circulated to all new hires and existing employees; each individual must acknowledge it in writing or via an online form.
- **Usage Monitoring:** The IT Department periodically reviews network and system logs for suspicious or excessive personal use. Violations are escalated as per HR guidelines.
- **Incident Handling:** If a misuse incident is detected, a formal investigation is initiated, and appropriate disciplinary measures (including revoking certain privileges) may be applied.
- **Protecting ISMS Data:** All logs and records regarding user behavior are labeled at least *Internal* and stored on secure servers. Only HR, the IT Security Team, and relevant department managers have access for investigative purposes.

Chapter 5

Risk Management Framework

This chapter outlines the core elements of GreenGrid Energy's Information Security Risk Management program. It includes the Risk Assessment Policy, the Risk Assessment and Treatment Processes, and a sample Inventory of Assets, developed in accordance with ISO/IEC 27001 Clause 6.1.2 and Annex A 5.9.

5.1 Risk Assessment Policy

Purpose and Scope

This policy establishes the approach for identifying, assessing, and treating information security risks at GreenGrid Energy. It applies to all systems, assets, processes, and personnel within the scope of the ISMS.

Objectives

- Provide a structured and repeatable method for risk identification and analysis.
- Support decisions on implementing adequate controls to protect information assets.
- Ensure compliance with legal, regulatory, and contractual requirements.

Governance and Risk Appetite

The organization has adopted a conservative risk appetite, especially for critical infrastructure such as SCADA systems and customer data. Risks categorized as *High* or *Critical* must be treated with priority and cannot be accepted without mitigation.

5.2 Risk Assessment Process

Overview

The risk assessment process consists of three main steps:

1. **Risk Identification:** Mapping assets, threats, vulnerabilities, and potential impacts.
2. **Risk Analysis:** Evaluating likelihood and consequence of threats exploiting vulnerabilities.

3. **Risk Evaluation:** Comparing identified risk levels against the organization's risk acceptance criteria.

Methodology

Risks are analyzed using a **Risk Level Matrix** that combines the likelihood of an event with the severity of its consequence. The matrix uses a 5x5 scale, where risk levels are categorized as Low, Medium, High, or Critical. This allows for a visual and systematic assessment.

		Impact →				
		Negligible	Minor	Moderate	Significant	Severe
Likelihood ↑	Very Likely	Low Med	Medium	Med Hi	High	High
	Likely	Low	Low Med	Medium	Med Hi	High
	Possible	Low	Low Med	Medium	Med Hi	Med Hi
	Unlikely	Low	Low Med	Low Med	Medium	Med Hi
	Very Unlikely	Low	Low	Low Med	Medium	Medium

Figure 5.1: Risk Level Matrix used by GreenGrid Energy

Source: gustavodefelice.com – What is a Risk Matrix?

The matrix helps determine which risks require immediate treatment and which may be accepted under certain conditions.

5.3 Risk Treatment Process

Treatment Options

Identified risks that exceed acceptable levels must be treated using one or more of the following strategies:

- **Mitigation:** Reducing risk through controls and safeguards.
- **Acceptance:** Acknowledging low risks with minimal impact.
- **Transference:** Outsourcing or insuring against the risk.
- **Avoidance:** Eliminating the risk by discontinuing risky activities.

Implementation

For each risk requiring treatment, a formal plan is created defining:

- The selected treatment strategy.
- Responsible stakeholders.
- Implementation timeline and monitoring procedures.

5.4 Inventory of Assets

A structured inventory of information assets is essential to support the organization's risk management and treatment activities. This section provides a description of four key assets considered relevant to the scope of the ISMS at GreenGrid Energy. Each asset is described according to its classification, use, ownership, governance responsibilities, and associated security concerns.

SCADA GridLink System

The SCADA GridLink system is a combination of embedded hardware and software used by the Engineering & Grid Operations department to perform real-time monitoring and control of the power grid infrastructure. It is classified as **Critical**, given its direct impact on grid stability and energy distribution. A compromise of this system could lead to widespread outages or unauthorized manipulation of power flows. Security measures include strict OT/IT network segmentation, physical access control, secure remote access protocols, and frequent firmware patching.

Governance: The Engineering & Grid Operations department is responsible for the asset's lifecycle, supported by the CISO and Security Operations team. Governance ensures alignment with national infrastructure protection regulations and internal operational continuity policies.

Compliance: Must comply with NIS2 Directive (EU) and ISO/IEC 27019 for energy sector control systems.

Microsoft 365 Suite

Microsoft 365 is a cloud-based productivity suite used by all departments across GreenGrid Energy for communication, document editing, cloud storage, and collaborative workflows. The asset is categorized as **Important**, due to its high usage and potential to expose internal documents and communications. The IT Department is responsible for its administration. Key risks include credential compromise through phishing attacks, unauthorized file sharing, and improper access control. The system is protected using Multi-Factor Authentication (MFA), conditional access policies, and centralized logging.

Governance: Managed by the IT Department in coordination with the ISMS team, ensuring secure configurations and incident response readiness.

Compliance: Must adhere to ISO/IEC 27001, 27017 (for cloud security), and internal Acceptable Use and Identity Management policies.

Customer Relationship Management (CRM) Database

The CRM Database, managed by the Customer Services & Billing department, stores personally identifiable information (PII), billing history, and service contracts. This asset is classified as **Highly Confidential**, as it involves sensitive customer data regulated by data protection laws such as the GDPR. Security measures include encryption at rest and in transit, strict role-based access control, regular auditing of access logs, and automated data retention enforcement.

Governance: Ownership lies with the Customer Services & Billing department, under oversight from the Data Protection Officer (DPO) and CISO for regulatory compliance and security posture.

Compliance: Must fully comply with the General Data Protection Regulation (GDPR), ISO/IEC 27701, and applicable data retention and breach notification laws.

SAP ERP System

The SAP ERP system is a business-critical software platform used primarily by the Supply Chain Management department for handling procurement, inventory control, invoicing, and vendor management. This asset is classified as **Important**, as it directly supports financial and operational continuity. The system processes sensitive financial data and operational records, which must be protected from unauthorized access, internal misuse, or misconfiguration.

Governance: Managed by the Supply Chain team in collaboration with the Enterprise Applications Unit within IT. The internal Audit and Compliance Office is responsible for ensuring traceability and segregation of duties.

Compliance: Subject to ISO/IEC 27001, and internal financial control requirements.

Classification and Prioritization Approach

All assets within the ISMS are evaluated using a classification model based on the Confidentiality, Integrity, and Availability (CIA) triad. The classification levels used include:

- **Critical** – Systems that must remain operational at all times to maintain business continuity.
- **Highly Confidential** – Assets containing sensitive or regulated information.
- **Important** – Supporting systems and tools that have a moderate to high impact if compromised.

Inventory Governance

Each asset has a designated asset owner and a responsible governance entity. The ISMS team coordinates with these asset owners to ensure that classification, protection, and lifecycle management are documented and reviewed periodically.

Governance Responsibilities:

- **SCADA GridLink:** Engineering & Grid Operations, with oversight from CISO and Regulatory Affairs.

- **Microsoft 365:** IT Department and ISMS Team.
- **CRM Database:** Customer Services & Billing, DPO, and CISO.
- **SAP ERP System:** Supply Chain Management, Internal Audit, and IT Enterprise Systems.

The asset inventory is maintained in a centralized repository and reviewed quarterly, or whenever significant changes occur in infrastructure, ownership, or regulatory obligations.

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